DFRobot DFR1154 Ai Camera Quick Start Guide

Objective:

The following guide is designed to "get you started" using the DFR1154 Ai Camera with Open Ai to identify objects with action prompts. A user friendly interface has been developed to assist you in capturing images, recording audio prompts, displaying results (both test and audio) and providing export tools.

What you need:

- DFR1154 - Ai Camera Module
- Arduino IDE 2.X
- USB A to USB C cable
- Optional
    - Ai Camera Tripod Kit
    - Arduino UNO
    - Mini Breadboard

Programming the DFR1154:

A user-friendly UI has been developed to assist you in using the DFR1154 it will need to be uploaded to the Ai camera prior to use.

- Standalone App – Zip File: open_ai_2_0_28
- Ai Automation App – Zip File: open_ai_automation (version 1.0.0)

Prior to programming the DFR1154 the following items need to be configured within the Arduino IDE:

Boards Manager URL:

- Under the "Preferences" tab enter in the Additional boards Manager URL the following address: ESP 32 Board Configration
    - A great tutorial on how to do this is found here: Random Nerds Tutorial

Boards Manager:

- Under the Tools Tab or Utility Sidebar find the Boards Manager
    - Under the search window look for ESP32
    - Find esp32 by Espressif
    - Install the board configuration

Note this can take a long time to install, if you get a timeout error, perform the following:

Select **File > Quit** from the Arduino IDE menus if it is running.
All Arduino IDE windows will close.

Open the file at the following path in any text editor:

If you are using Linux:

`/home/<username>/.arduinoIDE/arduino-cli.yaml`

(where `<username>` is your Linux username)

The `.arduinoIDE` folder may be hidden by default in your file manager and terminal.

If you are using macOS:

`/Users/<username>/.arduinoIDE/arduino-cli.yaml`

(where `<username>` is your macOS username)

The `.arduinoIDE` folder is hidden by default. You can make it visible by pressing the **`Command`** + **`Shift`** + **`.`** keyboard shortcut.

If you are using Windows:

`C:\Users\<username>\.arduinoIDE\arduino-cli.yaml`

(where `<username>` is your Windows username)

Add the following content to the `arduino-cli.yaml` file (or replace the existing content if equivalent content is already present in the file):

```
network:
connection_timeout: 600s
```

ⓘ I arbitrarily chose a timeout value of 600 seconds. You can adjust this as needed.

Save the file.

Start Arduino IDE.

Now try installing/updating the "**esp32**" boards platform again, just the same as you did before. Hopefully this time it will be successful.

ESP32 Configuration:

- Under tools select the following:
  - Board : “ESP32S3 Dev Module”
  - USB CDC On Boot: “Enabled”
  - Flash Size: “16MB (128Mb)”
  - Partition Scheme “16M Flash (3MB APP,/9.9MB FATFS)”
  - PSRAM: “OPI PSRAM”

Code Libraries:

- AdruinoJson by Benoit Blanchon – 7.42 or newer

Open Ai Certificate:

- Open Ai Certificate - Create an Account
- Once you have created an account download your personal API Key
- Save the key, you will use it later.
  - Make sure you do not alter the key in any way.

Code Configuration:

- Open in the Arduino the program open_ai_2_0_28
- Enter your wifi ssid = “YourSSID” (~line 46)
- Enter your wifi password = “YourPassword” (~line 47)
- Enter your OpenAI_API_Key = “sk-....”
- Save and compile – look/correct for any errors if they appear.

Run the Code:

- Hook up the DFR1154 to your computer and select the proper com port
- Upload the code to the DFR1154
- Open the serial monitor and look for the Web Browser IP:

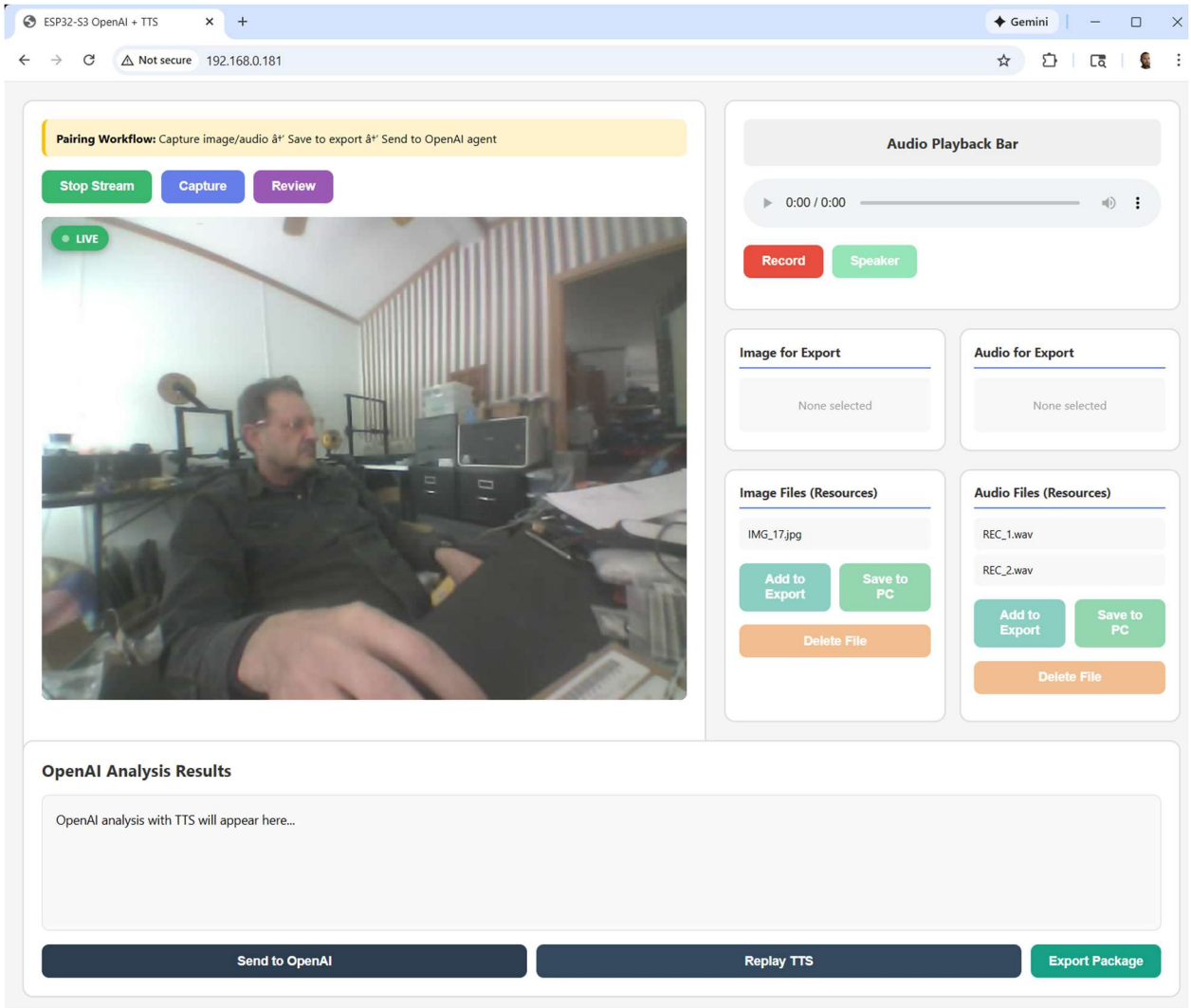
```
========================================
ESP32-S3 - OpenAI Interface v2.0.28
========================================

SD Card OK
Camera OK
PDM Microphone OK
Speaker OK
Connecting to WiFi...
WiFi connected
IP: 192.168.0.181

========================================
SYSTEM READY - Interface v2.0.28
Optimized pairing workflow
API Key status: Configured
========================================

Initialized counters - Images: 17, Audio: 2, TTS: 12
```

Open your web browser:

- Using the IP listed from the serial monitor open your web browser and paste in the URL (web browser address window).

- The results are shown below: